

The idea for this project is a chat application that provides assistance on site and then follows up with a phone call or email.

Project Name and one-sentence pitch

LoopChat — closes the loop between site visit and follow-up

An open-source AI agent that learns your website, engages visitors in real-time chat, and automatically follows up by email or phone — turning anonymous traffic into qualified leads.

What is the core problem the project aims to solve? Cover: (1) who the target user is, (2) the key pain points, (3) the specific tasks your AI Agent will execute.

Core Problem & Target Scenario

Healthcare providers — from independent clinics and specialist practices to telehealth platforms — lose prospective patients the moment they leave their website. A patient visits a clinic's site looking for answers about a procedure, insurance acceptance, or appointment availability, finds no immediate help, and moves on to the next provider. The clinic never knew they were there.

Target User

Small to mid-sized healthcare providers: independent medical practices, dental offices, mental health clinics, physical therapy centers, and telehealth platforms. Their front desk staff are already overwhelmed with in-person patients, phone calls, and insurance paperwork — they cannot also monitor a live chat queue or personally follow up with every website visitor.

Key Pain Points

Prospective patients have specific, sensitive questions (insurance coverage, procedure costs, provider credentials, wait times) that a static FAQ page cannot answer — so they leave without booking

Clinics have no visibility into who visited their site or what they were looking for

Patient intake is still largely paper-based or generic web forms that don't adapt to what the patient actually asked

Follow-up is slow and impersonal — by the time the front desk calls back, the patient has already booked elsewhere

HIPAA concerns make providers deeply uncomfortable routing patient conversations through third-party platforms they don't control — which is exactly why a self-hostable, open-source solution is a stronger fit here than Intercom or Drift

What the AI Agent Executes

Website ingestion — the agent crawls and indexes the clinic's website: services offered, providers, insurance accepted, locations, FAQs — building a knowledge base so it can answer prospective patient questions accurately and in context

Real-time chat engagement — an embeddable chat widget greets visitors, answers their specific questions using the clinic's own content, and identifies intent (booking a consultation, asking about a condition, checking insurance)

Structured intake capture — when a visitor expresses interest, the agent surfaces a lightweight, conversational intake form — collecting name, contact details, reason for visit, and insurance info — naturally, within the flow of the conversation

Personalized follow-up — based on the chat transcript and intake data, the agent automatically sends a personalized follow-up email or initiates a voice callback that references exactly what

the patient asked about, while their intent is still fresh — dramatically reducing the time between first touch and booked appointment

Why this matters in healthcare specifically

The average clinic loses 30–40% of prospective patients to competitors simply due to slow or absent follow-up. An AI agent that responds instantly, captures structured intake data, and follows up within minutes — without burdening front desk staff — directly improves patient acquisition while reducing administrative load. Because the system is open-source and self-hostable, clinics maintain full control over patient conversation data, making it a credible option for HIPAA-conscious environments.

Agent Data Flow & Workflow*

Briefly outline the Agent's data flow. Example: User input → Orchestrator → Call Search API → Fetch DB metadata → Generate Response → Save Session logs.

Agent Data Flow & Workflow

Phase 1 — Website Ingestion (One-time setup)

Clinic URL input → Web Crawler (Firecrawl) → Raw markdown content → Text chunker → Embedding model (Qwen/Alibaba Cloud) → Vector store (Qdrant) → Ingestion confirmation

Phase 2 — Real-Time Visitor Engagement (Per session)

Visitor opens chat widget → Session initialized → NeonDB logs session → Visitor message input → Orchestrator → Retrieve relevant chunks from Qdrant (RAG) → Qwen LLM generates context-aware response → Response delivered to chat widget → Session transcript updated in NeonDB → Intent classifier checks for booking/inquiry signals → If intent detected → Intake form surfaced to visitor → Visitor submits form (name, contact, insurance, reason for visit) → Lead record saved to NeonDB

Phase 3 — Personalized Follow-Up (Post-session)

Session ends or visitor exits → Orchestrator retrieves full transcript + intake data from NeonDB → Qwen LLM generates personalized follow-up message referencing visitor's specific questions →

Email path: Resend/SendGrid delivers personalized email to visitor within minutes

Voice path (stretch): Vapi/Bland.ai initiates AI voice callback to visitor's phone number

Logging & Observability

All agent actions → Session logs saved to NeonDB → Clinic dashboard displays leads, transcripts, follow-up status, and conversion metrics

Simplified linear view for the form:

Visitor lands on site → Chat widget loads → Visitor message → Orchestrator → RAG query (Qdrant) → Qwen LLM → Response to visitor → Intent detected → Intake form captured → Lead saved (NeonDB) → Follow-up triggered → Personalized email or voice call sent → Clinic dashboard updated

Minimum Viable Product (MVP) Definition*

What is the exact end-to-end working flow I should plan to demonstrate on Demo Day?
1-3 sentences.

A clinic representative pastes their website URL into the dashboard, and the system ingests and indexes the site's content in under 60 seconds. A prospective patient then interacts with the embeddable chat widget, asks questions about services or insurance, and completes a lightweight intake form — all powered by a Qwen-based RAG agent with full knowledge of the clinic's site. Within minutes of the conversation ending, the patient receives a personalized follow-up email referencing exactly what they asked about, while the clinic dashboard displays the captured lead, full transcript, and follow-up status in real time.

Three sentences, covers the full loop, and gives you three distinct visual moments for a live demo:

The URL ingestion (fast, satisfying, shows the tech)

The chat + intake flow (the core product experience)

The follow-up email arriving + dashboard updating (the "wow" closer)